ORACLE DATA GUARD 19c

Daily Monitoring Guide

Step-by-Step DBA Checklist

Primary & Standby Database Health | Redo Apply | Lag Monitoring | Switchover Readiness

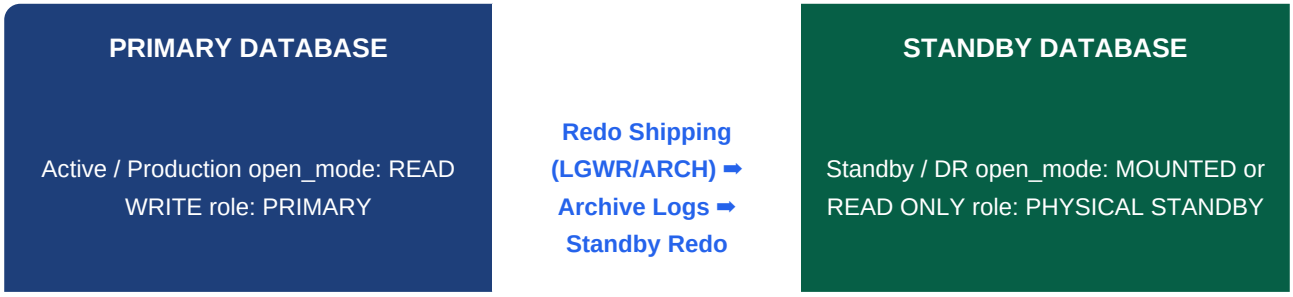
Prepared By	Designation	Document Date
Bidhan Mandal	Oracle DBA	March 09, 2026

Table of Contents

- Step 1 — Check Primary Database Status
- Step 2 — Check Standby Database Status
- Step 3 — Check Data Guard Configuration (dgmgrl)
- Step 4 — Monitor Redo Apply Lag
- Step 5 — Check Archive Log Gap
- Step 6 — Verify MRP / RFS Process Status
- Step 7 — Check Alert Log on Primary & Standby
- Step 8 — Verify Data Guard Broker Status
- Step 9 — Check Standby Redo Logs
- Step 10 — Test Switchover Readiness
- Step 11 — Monitor Redo Transport & Network
- Step 12 — Daily Data Guard Checklist Summary

Data Guard Architecture Overview

Oracle Data Guard maintains one or more synchronized copies (standby databases) of a production database. The Primary database ships redo data to Standby databases, which apply the redo to stay synchronized. Daily monitoring ensures zero data loss and fast failover capability.



INFO: *All SQL queries below should be run on BOTH Primary and Standby unless explicitly stated. Connect using: sqlplus / as sysdba*

STEP 1 — CHECK PRIMARY DATABASE STATUS

Verify the primary database is open, in ARCHIVELOG mode, and operating as PRIMARY role. This is the foundation of every daily Data Guard check.

```
-- 1.1 Check primary database role and status (run on PRIMARY)
SELECT name, db_unique_name, open_mode, log_mode,
       database_role, protection_mode, protection_level,
       switchover_status
FROM v$database;
```

```
-- 1.2 Check instance status
SELECT instance_name, host_name, version_full,
       startup_time, status, active_state
FROM v$instance;
```

HEALTHY IF: *open_mode=READ WRITE | database_role=PRIMARY | log_mode=ARCHIVELOG | switchover_status=TO STANDBY or SESSIONS ACTIVE*

Parameter	Expected Value	Action if Wrong
database_role	PRIMARY	Check DG config — may have switched over
open_mode	READ WRITE	Database may be mounted only — open it
protection_mode	MAXIMUM PERFORMANCE or AVAILABILITY	Verify DG protection mode config
switchover_status	TO STANDBY	If NOT ALLOWED — investigate DG errors

STEP 2 — CHECK STANDBY DATABASE STATUS

Verify the standby database is mounted or open in READ ONLY mode and actively receiving redo data from the primary.

```
-- 2.1 Check standby database role and status (run on STANDBY)
SELECT name, db_unique_name, open_mode, log_mode,
       database_role, protection_mode,
       switchover_status, dataguard_broker
FROM v$database;
```

```
-- 2.2 Check if redo apply (MRP) is running on standby
SELECT process, status, sequence#, delay_mins
FROM v$managed_standby
WHERE process IN ('MRP0','RFS','ARCH')
ORDER BY process;
```

```
-- 2.3 Check standby apply status detail
SELECT apply_mode, recovery_mode, standby_became_primary_scn
FROM v$standby_log_info;

-- 2.4 Check current SCN difference
SELECT current_scn FROM v$database;  -- Run on both Primary & Standby
```

HEALTHY IF: *database_role=PHYSICAL STANDBY | MRP0 status=APPLYING_LOG | open_mode=MOUNTED or READ ONLY WITH APPLY*

WARNING: *If MRP0 process is not running, redo is NOT being applied. Start it: ALTER DATABASE RECOVER MANAGED STANDBY DATABASE DISCONNECT FROM SESSION;*

STEP 3 — CHECK DATA GUARD CONFIGURATION (dgmgrl)

Use the Data Guard Broker command-line interface (dgmgrl) to get a complete view of your Data Guard configuration health and status in one command.

```
# 3.1 Connect to Data Guard Broker (run on Primary or Standby)
dgmgrl sys/password@PRIMARY_DB

# 3.2 Show full configuration summary
DGMGRL> SHOW CONFIGURATION;

# 3.3 Show detailed database status
DGMGRL> SHOW DATABASE VERBOSE primary_db;
DGMGRL> SHOW DATABASE VERBOSE standby_db;
```

```
# 3.4 Check for any warnings or errors
DGMGRL> SHOW CONFIGURATION LAG;

# 3.5 Validate the full configuration
DGMGRL> VALIDATE DATABASE primary_db;
DGMGRL> VALIDATE DATABASE standby_db;

# 3.6 Show all properties
DGMGRL> SHOW DATABASE PRIMARY_DB LogXptMode;
DGMGRL> SHOW DATABASE STANDBY_DB DelayMins;
```

dgmgrl Status	Meaning	Action Required
SUCCESS	All is well	No action needed
WARNING	Minor issue detected	Review and fix warning
ERROR	Configuration problem	Investigate immediately
DISABLED	Broker not managing this DB	Enable broker management
CONNECT TIMEOUT	Cannot reach standby	Check network/listener

STEP 4 — MONITOR REDO APPLY LAG

Apply lag is the most critical daily metric in Data Guard. It shows how far behind the standby is from the primary. High lag means more data loss in a failover scenario.

```
-- 4.1 Check apply lag (run on STANDBY)
SELECT name, value, datum_time
FROM v$dataguard_stats
WHERE name IN ('transport lag','apply lag','apply finish time',
               'estimated startup time');
```

```
-- 4.2 Check lag via v$standby_event_histogram
SELECT apply_lag, count
FROM v$standby_event_histogram
WHERE count > 0
ORDER BY apply_lag;
```

```
-- 4.3 Check lag via Data Guard Broker
DGMGRL> SHOW CONFIGURATION LAG;

-- 4.4 Check lag using v$managed_standby (MRP process)
SELECT process, status, client_process,
       sequence#, block#,
       ROUND(blocks*512/1048576,1) AS applied_mb
FROM v$managed_standby
WHERE process = 'MRP0';
```

Apply Lag	Status	Threshold	Action
0 – 30 sec	Excellent	Normal	No action needed
30s – 5 min	Good	Acceptable	Monitor — may be network burst
5 – 30 min	Warning	Investigate	Check network, MRP, archive delivery
30 min – 1hr	Critical	Escalate	Notify management, investigate root cause
> 1 hour	EMERGENCY	URGENT	Immediate action — DG at risk

STEP 5 — CHECK ARCHIVE LOG GAP

An archive log gap means the standby is missing archive logs that need to be applied. This must be resolved immediately to prevent increasing data loss exposure.

```
-- 5.1 Check for archive gap (run on PRIMARY)
SELECT thread#, low_sequence#, high_sequence#
FROM v$archive_gap;

-- If no rows returned: NO ARCHIVE GAP EXISTS (Good!)
```

```
-- 5.2 Check sequence numbers on Primary vs Standby

-- On PRIMARY:
SELECT thread#, sequence#, archived, applied
FROM v$archived_log
WHERE first_time >= SYSDATE - 1
ORDER BY thread#, sequence# DESC
FETCH FIRST 20 ROWS ONLY;

-- On STANDBY:
SELECT thread#, sequence#, applied
FROM v$archived_log
WHERE first_time >= SYSDATE - 1
ORDER BY thread#, sequence# DESC
FETCH FIRST 20 ROWS ONLY;
```

```
-- 5.3 Manually register missing archive log on Standby (if gap found)
ALTER DATABASE REGISTER LOGFILE '/path/to/archive_log.arc';

-- 5.4 Copy missing archive from Primary to Standby (OS command)
scp primary:/archive/1_52_1050234567.dbf standby:/archive/
```

CRITICAL: *A sequence gap that keeps growing means automatic gap resolution (FAL_SERVER) is not working. Check FAL_SERVER and FAL_CLIENT parameters.*

STEP 6 — VERIFY MRP / RFS PROCESS STATUS

MRP (Managed Recovery Process) applies redo on the standby. RFS (Remote File Server) receives redo from the primary. Both must be running for Data Guard to function.

```
-- 6.1 Check all Data Guard processes (run on STANDBY)
SELECT process, pid, status, client_process,
       thread#, sequence#, delay_mins
FROM v$managed_standby
ORDER BY process;
```

Process	Role	Expected Status	If Missing/Wrong
MRP0	Managed Recovery (apply redo)	APPLYING_LOG	Start MRP: ALTER DATABASE RECOVER MANAGED STANDBY DATABASE DISCONNECT;
RFS	Remote File Server (receive)	IDLE / WRITING	Check Primary LGWR/ARCH, check network
ARCH	Archiver (archive redo)	CONNECTED	Check archiving on Primary
LNS	Log Network Server (Primary)	WRITING	Check on Primary — network issue

```
-- 6.2 Stop and restart MRP if needed (STANDBY)
-- Stop MRP:
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE CANCEL;

-- Start MRP (background, disconnected):
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE
  DISCONNECT FROM SESSION;

-- Start MRP with Real-Time Apply:
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE
  USING CURRENT LOGFILE DISCONNECT FROM SESSION;
```

STEP 7 — CHECK ALERT LOG ON PRIMARY & STANDBY

Always check alert logs on BOTH primary and standby every morning. Data Guard errors appear in the alert log before they become critical problems.

```
# 7.1 Get alert log path (on each server)
sqlplus / as sysdba
SELECT value FROM v$diag_info WHERE name = 'Diag Trace';
```

```
# 7.2 Check last 100 lines of alert log
tail -100 $ORACLE_BASE/diag/rdbms///trace/alert_.log

# 7.3 Filter Data Guard specific errors
grep -i "ORA-\|MRP\|RFS\|FAL\|GAP\|ERROR\|STANDBY\|APPLY" \
  $ORACLE_BASE/diag/rdbms///trace/alert_.log \
  | tail -50
```

```
-- 7.4 Query alert log via Oracle view (19c)
SELECT originating_timestamp, message_text
FROM v$diag_alert_ext
WHERE originating_timestamp >= SYSDATE - 1
  AND (message_text LIKE '%ORA-%'
    OR message_text LIKE '%MRP%'
    OR message_text LIKE '%GAP%'
    OR message_text LIKE '%apply%')
ORDER BY originating_timestamp DESC
FETCH FIRST 40 ROWS ONLY;
```

Error / Message	Meaning	Action
ORA-00600	Internal Oracle error	SR to Oracle Support immediately
ORA-01031	Insufficient privileges	Check DG user permissions
ORA-16191	Primary log shipping offline	Check network, restart LGWR
ORA-16401	archivelog rejected by standby	Check standby disk space
FAL: request gap	Archive gap detected	Check FAL_SERVER setting
LGWR: Error 3113	TNS lost contact with standby	Check network & standby listener

STEP 8 — VERIFY DATA GUARD BROKER STATUS

The Data Guard Broker automates and monitors DG configurations. A daily broker health check reveals issues with redo transport, apply services, and overall configuration integrity.

```
# 8.1 Full broker status check
dgmgrl / as sysdba

DGMGRL> SHOW CONFIGURATION;
-- Expected output:
-- Configuration - dg_config
--   Protection Mode: MaxPerformance
--   Members:
--   PRIMARY_DB  - Primary database
--   STANDBY_DB  - Physical standby database
-- Fast-Start Failover:  Disabled
-- Configuration Status: SUCCESS
```

```
# 8.2 Show full database details
DGMGRL> SHOW DATABASE VERBOSE STANDBY_DB;

# 8.3 Check health of both databases
DGMGRL> VALIDATE DATABASE PRIMARY_DB;
DGMGRL> VALIDATE DATABASE STANDBY_DB;

# 8.4 Check Fast-Start Failover status
DGMGRL> SHOW FAST_START FAILOVER;

# 8.5 Fix a disabled configuration
DGMGRL> ENABLE CONFIGURATION;
DGMGRL> ENABLE DATABASE STANDBY_DB;
```

```
-- 8.6 Check broker via SQL (alternative)
SELECT name, role, enabled, db_unique_name,
       protection_mode, fast_start_failover_status
FROM v$dg_broker_config;
```

STEP 9 — CHECK STANDBY REDO LOGS

Standby Redo Logs (SRLs) are required for Real-Time Apply and Maximum Availability/Protection modes. They must be present on the standby and properly configured.

```
-- 9.1 Check Standby Redo Logs on STANDBY
SELECT group#, thread#, sequence#, archived, status,
       ROUND(bytes/1048576) AS size_mb
FROM v$standby_log
ORDER BY group#;
```

```
-- 9.2 Check Online Redo Logs size (to match SRL size)
SELECT group#, members, status,
       ROUND(bytes/1048576) AS size_mb
FROM v$log
ORDER BY group#;

-- 9.3 Check that SRL count = ORL count + 1 per thread
-- e.g. if Primary has 3 Online Redo Log groups,
-- Standby should have at least 4 Standby Redo Log groups
```

```
-- 9.4 Add Standby Redo Logs if missing (run on STANDBY)
ALTER DATABASE ADD STANDBY LOGFILE THREAD 1
  '/u01/app/oracle/oradata/STBY/srl01.log' SIZE 200M;
ALTER DATABASE ADD STANDBY LOGFILE THREAD 1
  '/u01/app/oracle/oradata/STBY/srl02.log' SIZE 200M;
ALTER DATABASE ADD STANDBY LOGFILE THREAD 1
  '/u01/app/oracle/oradata/STBY/srl03.log' SIZE 200M;
ALTER DATABASE ADD STANDBY LOGFILE THREAD 1
  '/u01/app/oracle/oradata/STBY/srl04.log' SIZE 200M;
```

INFO: *Standby Redo Logs must be the same size as or larger than the Online Redo Logs on the Primary. Number of SRLs = (Number of ORL groups × threads) + 1.*

STEP 10 — TEST SWITCHOVER READINESS

Verify every day that the database is ready for a switchover or failover. Being "ready" means no data will be lost and the switchover can complete quickly.

```
-- 10.1 Check switchover status (run on PRIMARY)
SELECT switchover_status FROM v$database;

-- Good values: TO STANDBY, SESSIONS ACTIVE
-- Bad values:  NOT ALLOWED, FAILED DESTINATION
```

```
-- 10.2 Check switchover status (run on STANDBY)
SELECT switchover_status FROM v$database;

-- Good values: NOT ALLOWED (normal), TO PRIMARY
-- Bad values:  FAILED DESTINATION
```

```
# 10.3 Validate switchover via Broker (NON-DISRUPTIVE TEST)
dgmgrl / as sysdba
DGMGRL> VALIDATE DATABASE VERBOSE STANDBY_DB;

# 10.4 Perform actual Switchover (ONLY when needed)
# Step 1 – Convert Primary to Standby:
DGMGRL> SWITCHOVER TO STANDBY_DB;
# The Broker handles the rest automatically
```

switchover_status	Meaning	Ready?
TO STANDBY	Primary ready to switch	YES
SESSIONS ACTIVE	Active sessions — still can switch	YES (with WAIT)
NOT ALLOWED	Standby not ready	NO — investigate
FAILED DESTINATION	Cannot reach standby	NO — fix network/redo
PREPARING	Switchover in progress	IN PROGRESS

STEP 11 — MONITOR REDO TRANSPORT & NETWORK

Redo transport depends on a stable network connection between primary and standby. Monitor transport errors and network latency daily to prevent silent lag accumulation.

```
-- 11.1 Check redo transport errors (run on PRIMARY)
SELECT dest_id, dest_name, status, target,
       archiver, schedule, destination,
       error, db_unique_name
FROM v$archive_dest
WHERE target = 'STANDBY'
  AND status != 'INACTIVE';
```

```
-- 11.2 Check redo transport status in detail
SELECT dest_id, error, status,
       TO_CHAR(completion_time,'DD-MON HH24:MI') AS last_archive
FROM v$archive_dest_status
WHERE target = 'STANDBY';
```

```
-- 11.3 Check network transmission speed (bytes/sec)
SELECT dest_id,
       ROUND(net_timeout,1)     AS timeout_sec,
       ROUND(transmit_speed,1)  AS transmit_kbps
FROM v$redo_transport_stats;

-- 11.4 Check DG transport mode
SELECT dest_id, archiver, transmit_mode
FROM v$archive_dest
WHERE target = 'STANDBY' AND status = 'VALID';
```

```
# 11.5 Test network connectivity to standby (OS)
tnsping STANDBY_DB

# Check listener on standby server
lsnrctl status

# Test TNS connection
sqlplus sys/password@STANDBY_DB as sysdba
```

WARNING: *If v$archive_dest shows status=ERROR, redo transport has stopped. Run: ALTER SYSTEM SET LOG_ARCHIVE_DEST_STATE_2=DEFER then ENABLE to reset the destination.*

Daily Data Guard Monitoring Checklist

Print this checklist and complete it every morning. File daily reports for audit and compliance purposes.

	Daily Check	SQL / Tool	Expected	Result
☐	Primary DB Role & Status	`v$database`	PRIMARY / READ WRITE	_____
☐	Standby DB Role & MRP Status	`v$managed_standby`	MOUNTED / APPLYING	_____
☐	dgmgrl SHOW CONFIGURATION	`dgmgrl`	SUCCESS	_____
☐	Apply Lag < 30 seconds	`v$dataguard_stats`	< 30 sec	_____
☐	No Archive Log Gap	`v$archive_gap`	0 rows	_____
☐	MRP0 & RFS Processes Running	`v$managed_standby`	APPLYING_LOG	_____
☐	Alert Log — No ORA- Errors	`v$diag_alert_ext`	No errors	_____
☐	Broker Configuration SUCCESS	`dgmgrl VALIDATE`	SUCCESS	_____
☐	Standby Redo Logs Present	`v$standby_log`	UNASSIGNED/ACTIVE	_____
☐	Switchover Status Ready	`v$database`	TO STANDBY	_____
☐	Redo Transport No Errors	`v$archive_dest`	VALID / NO ERROR	_____

Date: _____ DBA Name: Bidhan Mandal Primary DB: _____ Standby DB: _____

Issues Found / Actions Taken:

Bidhan Mandal | Oracle Database Administrator | Oracle Data Guard 19c | March 09, 2026